

DecodeLabs – Data Analytics Project 3

SQL Data Analysis – Completed Project

Basis: The supplied Project 3 PDF defines the goal as using SQL queries to extract insights from a dataset and explicitly requires SELECT, WHERE, ORDER BY, GROUP BY, and basic aggregations COUNT, SUM, and AVG.

Dataset: Dataset 3.xlsx

Records: 1200 | **Columns:** 14

1. Dataset Structure

Column	Purpose / Type
OrderID	Order identifier
Date	Order date
CustomerID	Customer identifier
Product	Product category
Quantity	Number of units
UnitPrice	Price per unit
ShippingAddress	Shipping address
PaymentMethod	Payment method
OrderStatus	Order status
TrackingNumber	Shipment tracking number
ItemsInCart	Items in customer's cart
CouponCode	Coupon code
ReferralSource	Referral source
TotalPrice	Total order value

2. Required SQL Analysis

The following queries directly demonstrate every SQL feature listed in the supplied Project 3 PDF.

SELECT – view all records

```
SELECT * FROM sales_orders;
```

SELECT – choose columns

```
SELECT OrderID, Date, Product, Quantity, UnitPrice, TotalPrice  
FROM sales_orders;
```

WHERE – filter quantity

```
SELECT OrderID, Product, Quantity, TotalPrice  
FROM sales_orders  
WHERE Quantity > 3;
```

ORDER BY – highest sales first

```
SELECT OrderID, Product, Quantity, TotalPrice  
FROM sales_orders  
ORDER BY TotalPrice DESC;
```

COUNT – total orders

```
SELECT COUNT(*) AS total_orders
```

```
FROM sales_orders;
```

GROUP BY + COUNT – orders by product

```
SELECT Product, COUNT(*) AS order_count  
FROM sales_orders  
GROUP BY Product  
ORDER BY order_count DESC;
```

GROUP BY + SUM – quantity by product

```
SELECT Product, SUM(Quantity) AS total_quantity  
FROM sales_orders  
GROUP BY Product  
ORDER BY total_quantity DESC;
```

GROUP BY + SUM – sales by product

```
SELECT Product, SUM(TotalPrice) AS total_sales  
FROM sales_orders  
GROUP BY Product  
ORDER BY total_sales DESC;
```

GROUP BY + AVG – average order value

```
SELECT Product, AVG(TotalPrice) AS average_order_value  
FROM sales_orders  
GROUP BY Product  
ORDER BY average_order_value DESC;
```

3. Analysis Results from Dataset 3

Product	Orders	Quantity	Total Sales	Avg Order
Chair	178	562	195,620.11	1,098.99
Printer	181	542	195,612.61	1,080.73
Laptop	173	535	192,126.56	1,110.56
Tablet	179	497	186,568.95	1,042.28
Monitor	163	480	175,651.41	1,077.62
Desk	170	508	167,459.93	985.06
Phone	156	411	151,722.39	972.58

Order Status Summary

Order Status	Orders	Total Sales
Cancelled	250	276,396.21
Returned	247	243,277.70
Pending	237	256,328.15
Shipped	235	246,159.58
Delivered	231	242,600.32

Payment Method Summary

Payment Method	Orders	Total Sales
Online	258	262,442.94
Cash	246	259,786.29
Credit Card	234	263,847.63
Debit Card	232	232,361.18
Gift Card	230	246,323.92

4. Top 10 Orders by Total Price

OrderID	Product	Qty	Unit Price	Total Price	Status
ORD200789	Tablet	5	691.28	3,456.40	Delivered
ORD201122	Monitor	5	678.19	3,390.95	Returned
ORD200632	Laptop	5	678.16	3,390.80	Delivered
ORD200469	Chair	5	676.98	3,384.90	Cancelled
ORD200328	Tablet	5	674.04	3,370.20	Cancelled
ORD200107	Printer	5	670.75	3,353.75	Shipped
ORD200326	Laptop	5	670.48	3,352.40	Returned
ORD201065	Printer	5	666.80	3,334.00	Delivered
ORD201031	Phone	5	664.51	3,322.55	Pending
ORD200463	Laptop	5	662.78	3,313.90	Shipped

5. SQL File

A separate PostgreSQL-compatible SQL script is included with this project. It contains the required SELECT, WHERE, ORDER BY, GROUP BY, COUNT, SUM, and AVG queries.

6. Conclusion

This project demonstrates SQL-based data analysis on the supplied Dataset 3.xlsx file. The analysis uses filtering, sorting, grouping, and aggregation to turn raw order data into useful business insights. These techniques match the core requirements stated in the supplied DecodeLabs Project 3 PDF.